

Arshad Mahmood

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About

Full-Stack Software Engineer with expertise in Angular, Java Spring Boot, Python, and AWS, specializing in AI-driven, scalable enterprise applications. Unique ability to blend engineering knowledge with software development, proven by impactful projects in water treatment, cost estimation platforms, and AI assistants. Recognized as Best Performer of the Year 2024 and 1st Rank in Veolia Techathon 2024 for driving digital transformation and AI solutions.

Education

Institute of Engineering and Technology , Lucknow
Bachelor of Technology in Chemical Engineering

June 2019 – June 2023

- CGPA: 8.11/10

Experience

Software Engineer

Bangalore, India

Veolia Water technologies and solutions

July 2024 – Present

- Built a Cooling Tower Water Treatment Calculator using Angular 16+ with complex reactive forms and heat map visualizations.
- Developed Tank Sizing & Cost Estimation Platform supporting 10+ concurrent configurations and location-based cost optimization.
- Integrated with Laboratory Information Management Systems (LIMS) for real-time water chemistry analysis.
- Revamped AI assistant reporting modules with intuitive UI and scalable architecture.
- Delivered end-to-end solutions used by 500+ employees, improving efficiency across business units.
- Collaborated cross-functionally in Agile teams to implement maintainable and scalable systems.

Graduate Engineer Trainee (GET)

Bangalore, India

Veolia Water technologies and solutions

July 2023 – July 2024

- Developed a Chemical Compatibility Tool using Spring Boot + SQL, combining chemical engineering principles with programming.
- Earned appreciation for creating innovative, efficient in-house tools that reduced dependency on external vendors.
- Strengthened understanding of water treatment technologies and engineering workflows.

Intern

Bangalore, India

Suez Water Technologies and Solutions

July 2022 – Sept 2022

- Automated reporting workflows and reference lists, improving operational efficiency.
- Gained exposure to Power BI, corporate networking, and cross-team collaboration.

Projects

Cooling Tower Water Treatment Program

2025

- **Tech Stack:** Angular 16+ (Standalone Components, Reactive Forms, RxJS), PrimeNG, Bootstrap, NGX-Translate, TypeScript, SCSS, RESTful APIs, JSON, JWT Authentication
- Designed and developed a responsive calculator that optimizes chemical treatment programs based on water chemistry analysis and system parameters.
- Designed intuitive heat map visualizations to display the effectiveness of treatment.
- Integrated with LIMS to import water chemistry data, and built a real-time API-based product selection tool.
- Developed JSON-based import/export functionality for collaboration and record-keeping.

AI-Powered Learning Assistant Development

- **Tech Stack:** Flask, Python, Google Gemini AI API, JavaScript

- Built a personalized AI learning platform with adaptive quizzes, structured prompt engineering, and progress analytics.
- Integrated Google Gemini 1.5 Flash API with structured prompt engineering and implemented robust fallback mechanisms for seamless user experience.
- Designed RESTful Flask backend with comprehensive error handling, environment management, and CORS configuration for cross-origin requests.

AI Assistant UI Development

- **Tech Stack:** Angular, RxJS, AWS (Cognito, CloudFront, S3, Lambda, CodeCommit)
- Led a UI redesign & integration with Insight, improving adoption and accessibility.
- Managed end-to-end deployment with CI/CD pipelines and ongoing product improvements.
- Managing the chatbot as the Product Owner, actively collaborating with stakeholders to define future enhancements, features, and expansion strategies company products.

Tank Sizing & Cost Estimation Platform

- **Tech Stack:** Angular, Ant Design, Google Sheets API
- Designed and developed a comprehensive tank sizing calculator that automates complex chemical storage calculations and cost estimations based on multiple parameters.
- Built dynamic form system supporting 10+ concurrent tank configurations with conditional rendering based on regional requirements.
- Added location-based features to automatically find nearest warehouses and calculate shipping costs for customers.
- Developed multi-format export system (PDF, Excel, JSON) with Google Sheets integration for quote generation.
- Implemented responsive design with advanced search functionality and server-side filtering for product catalogs.

Reporting Module Revamp

2024

- **Tech Stack:** Angular, Angular Material, PrimeNG, AWS
- Collaborated across teams to design and deliver a scalable, user-friendly reporting module from scratch.
- Incorporated feedback and continuously improved functionality.

Attendance Monitoring Tool

2024

- **Tech Stack:** App Script, HTML, CSS, Google Sheets
- Built a tool for 500+ users, streamlining attendance tracking across locations.

Chemical Compatibility Tool

2023

- **Tech Stack:** HTML, CSS, JavaScript, Java Spring Boot, SQL
- Sole developer; integrated chemical engineering with programming.
- Established a strong foundation for future internal full-stack tools.

Skills & Technologies

Programming: Java, Python, JavaScript (ES6+), TypeScript, SQL, HTML, CSS, AppScript

Frameworks: Angular, Spring Boot, Flask, RxJS, PrimeNG, Angular Material, Bootstrap

Cloud & Tools: AWS (S3, Lambda, CloudFront, Cognito, CodeCommit), Power BI, JIRA, Google Apps Script

Practices: Agile, REST API Design, AI Integration, CI/CD Pipelines, Data Visualization

Soft Skills: Problem-Solving, Teamwork, Communication, Leadership, Innovation

Awards

- Best Performer of the Year 2024
- 1st Rank in Veolia Techathon 2024

Languages

English, Hindi